Measuring Muscle Fatigue With Electromyography

Melvin Sassani, Adom Mia, Aziz Aytbayev, Assem Taleb

Abstract

Muscle fatigue can be difficult to measure due to the wave signal. Due to being generated by an EMG-sensor, it is not always accurate and therefore can be challenging to read. As such, obtaining correct data from the electric signal generated by the muscle is crucial for the determination of muscle fatigue. This project aims to collect as much data as possible and develop a pattern recognition system that can determine if the muscle is exhibiting signs of fatigue. The expected result is to identify a measurable pattern in the EMG signal that can distinguish between a fatigued and non fatigued muscle.

Introduction

Surface electromyography (sEMG) can be used to evaluate muscle fatigue by measuring the electrical activity of the muscle. However, the signal is affected by factors such as noise, sensor properties and electrode placement. Therefore making reliable signal acquisition a priority before fatigue can be analyzed [1], [2]. Previous research indicates that muscle fatigue can be identified by analyzing features of the sEMG signal. Fuentes del Toro et al. used wavelet filtering and evaluated RMS (Root Mean Square), MAV (Mean Absolute Value), and MNF (Mean Frequency). Showing that RMS and MAV increased while MNF decreased as fatigue developed [3].

This project aims to evaluate an sEMG measurement system that amplifies, filters, and records muscle activity. The recorded data is analyzed over time using RMS, MAV, and MNF to investigate whether a consistent fatigue related pattern can be identified.

Design

This project presents a wearable sEMG measurement system consisting of three electrodes with a signal processing device as shown in figure 2. The electrodes detect the electric signals generated by the muscle, which are then amplified and filtered using an analog circuit powered by two 9V batteries. An ESP32 converts the conditioned analog signal to a digital signal by using its built-in ADC, which is controlled by C++ code. The acquired data can then be transferred via WIFI to a computer, where a python program calculates RMS and MAV and then plots the results for analysis of muscle activation and fatigue[figure 1].

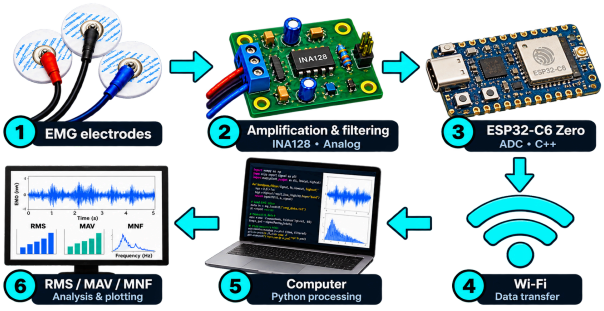


Figure 1: System overview of the signal flow from sEMG electrodes to signal processing, data transfer and result visualization (Image generated by AI).

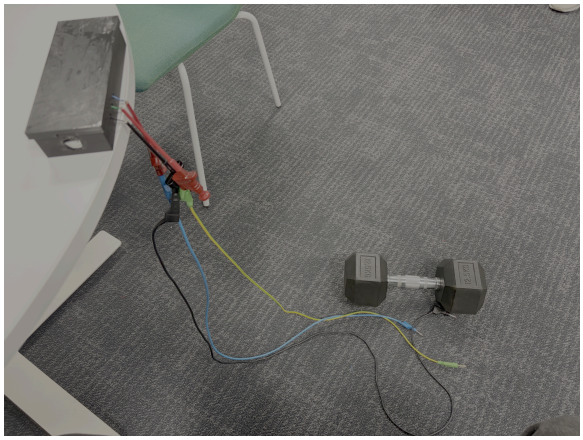


Figure 2: The developed sEMG measurement system, with all electronic components housed inside the black box. The measurement electrodes are connected via the blue and yellow cables, while the reference electrode is connected via the black cable.

Evaluation

The system successfully recorded clear signals of muscle activity during repeated biceps curls with a 10 kg weight. As shown in figure 3, the signal amplitude generally increased as more repetitions were performed. A similar pattern was observed across the other tests, suggesting a consistent trend within the collected measurements. The participant also reported feeling progressively more fatigued as the amplitude increased, which supported the observed relationship between perceived fatigue and signal amplitude.

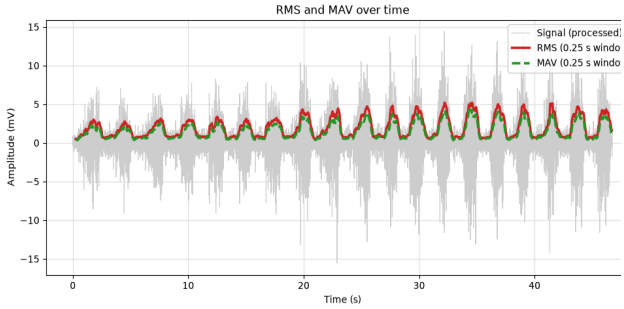


Figure 3: Processed sEMG signal (grey) with RMS (red) and MAV (green dashed), shown over a 50 seconds window. The curves show changes in signal amplitude over time during repeated muscle contraction.

The recorded data were processed using a Python program on the computer, which calculated RMS and MAV over a 0.25 second window. These measurements made changes in signal amplitude easier to follow throughout the exercise.

However, the amount of collected data was insufficient to establish a reliable threshold for determining whether the muscle was fatigued. Although the results were promising, increased amplitude alone could not confirm fatigue. Further measurements across multiple participants and testing conditions are needed to evaluate this relationship.

Conclusion

This project demonstrated that the developed sEMG system could record clear muscle activity signals and calculate RMS and MAV using a Python program. During repeated biceps curls with 10 kg weight, signal amplitude generally increased alongside reported fatigue. Similar patterns were observed across other tests. However, the collected data were insufficient to establish a reliable fatigue threshold. Future work should include more testing with participants and more evaluation of the data. The device will also be made lighter and more compact. An adjustable elastic band will wrap around the limb and hold the electrodes in place over the muscle being measured. The box containing all electronic components will be attached to the same band, allowing the user to wear the complete system during exercise.

References

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